

Script-Controlled ACL – Restrict Record Access Based on Field Value

Group Project:

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Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

Skill Rquired

ServiceNow

Scripting

Access Control List

Milestone-1 Creation of Users and Roles

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**

- Navigate to **User Administration** → **Users**
- Click **New**
- **User ID:** EEE User
- **First name:** EEE
- **Last name:** User
- **Email:** eeeuser@gmail.com
- Save the record

User Administration - User: EEE User

Buttons: Update, Set Password, Delete

Fields:

- User ID: EEE User
- First name: EEE
- Last name: User
- Title:
- Department:
- Language: -- None --
- Calendar integration: Outlook
- Time zone: System (America/Los_Angeles)
- Date format: System (yyyy-MM-dd)
- Business phone:
- Mobile phone:
- Photo: Click to add...
- Active: ☒
- Identity type: Human
- Internal Integration User: ☐
- Password needs reset: ☐
- Locked out: ☐

Buttons: Update, Set Password, Delete

Role Creation

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

Role Administration - Role: bb1

Buttons: Update, Delete

Fields:

- Name: bb1
- Application: Global
- Elevated privilege: ☐
- Description:

Buttons: Update, Delete

Tabs: Contains Roles, Applications with Role, Modules with Role, Custom Tables

servicenow - Role: bb2

Buttons: Update, Delete

Fields:

- Name: bb2
- Application: Global
- Elevated privilege: ☐
- Description:

Buttons: Update, Delete

The screenshot shows the ServiceNow Roles form for role 'bb3'. The header includes the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin, and Roles. A search bar is present. The form fields include: Name (bb3), Application (Global), Elevated privilege (checkbox), and Description (text area). At the bottom, there are 'Update' and 'Delete' buttons.

The screenshot shows the ServiceNow Roles form for role 'bb4'. The header is similar to the previous one, but the 'Roles' tab is highlighted. The form fields include: Name (bb4), Application (Global), Elevated privilege (checkbox), and Description (text area). At the bottom, there are 'Update' and 'Delete' buttons.

Tables Creation

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false

The screenshot shows the ServiceNow Table form for 'Institution Details'. The header includes the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin, and Tables. A search bar is present. The form fields include: Label (Institution Details), Name (u_institution_details), Application (Global), and Remote Table (checkbox). At the bottom, there are 'Delete', 'Update', and 'Delete All Records' buttons. A message bar at the top states: 'This form has annotations - click ⓘ to toggle them - (Click here to never show this again)'.

3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number** – Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email** – String
 - **Phone Number** – String
 - **Description** – Multi String
5. Save the Table

Table Institution Details

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Updates	Integer	(empty)		40	false
Branch	Choice	(empty)		40	false
Updated	Date/Time	(empty)		40	false
Phone Number	String	(empty)		40	false
Student Roll Number	String	(empty)		40	false
Faculty Name	Reference	User		32	false
Created by	String	(empty)		40	false
Email	String	(empty)		40	false
Description	String	(empty)		40	false
Created	Date/Time	(empty)		40	false
Student Name	Reference	User		32	false

Insert a new row...

No templates are available Create A New One?

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

Institution Details Branch Search

Actions on selected rows... New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE		allan@gmail.com	Alfonso Griglen	989076	Allan Schwantd	STU500
CSE		andrew@gmail.com	Allie Pumphrey	897654	Andrew Och	STU300
EEE		allan@gmail.com	Alejandro Mascall	43543556	Allan Schwantd	STU400
ECE		aileen@gmail.com	Abel Tuter	231324	Aileen Mottern	STU100
EEE		alejandro@gmail.com	Abraham Lincoln	546738	Alejandro Mascall	STU200

Creation of Access Control Lists(Read ACL)

1. In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
2. Elevate "Security_admin" role
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
 - **Advanced:** true
5. In the **Requires role** related list, add:
 - bb1 (custom role)
6. Add Data condition as **Branch is EEE**

7. In the **Script** field, add the following script:

```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  // Allow only EEE branch users to see EEE records  
  if (gs.hasRole('bb1')){  
    return true;  
  }  
  // Deny access for all others  
  return false;  
})();
```

8. Click **Submit** to save the ACL.

Elevate role

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update

Access Control
u_institution_details

* Type

record

* Operation

read

Decision Type

Allow If

Admin overrides

☒

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Applies To

Add Filter Condition

Add OR Clause

Application

Global

Active

☒

Advanced

☒

Update

Delete

Conditions

Requires role

Role

bb1

Security Attribute Condition

Access Control u_institution_details

Data Condition

Condition No. of records matching the condition: 1

Add Filter Condition Add OR Clause

Branch is EEE

AND OR

Advanced Condition

Controlled by References

Script Turn on ECMAScript 2021 (ES12) mode

```

1 (function() {
2   if (gs.hasRole('admin')) {
3     return true;
4   }
5
6   if (gs.hasRole('bb1')) {
7     return true;
8   }
9
10  return false;
11})();
12

```

9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

- User with **bb1** role can view **only EEE branch records**

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL is excuted	allan@gmail.com	Alfonso Griglen	989076	Allan Schwantd	STU500
EEE	ACL is excuted	allan@gmail.com	Alejandro Mascall	43543556	Allan Schwantd	STU400
EEE	ACL is excuted	alejandro@gmail.com	Abraham Lincoln	546738	Alejandro Mascall	STU200

- User without the role cannot view any records
- Admin user can view **all records regardless of branch**

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE		allan@gmail.com	Alfonso Griglen	989076	Allan Schwantd	STU500
CSE		andrew@gmail.com	Allie Pumphrey	897654	Andrew Och	STU300
EEE		allan@gmail.com	Alejandro Mascall	43543556	Allan Schwantd	STU400
ECE		aileen@gmail.com	Abel Tuter	231324	Aileen Mottern	STU100
EEE		alejandro@gmail.com	Abraham Lincoln	546738	Alejandro Mascall	STU200

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

The screenshot shows the ServiceNow 'Access Control' form for the record 'u_institution_details'. The form is titled 'Access Control - u_institution_details' and includes a search bar and navigation links. The form fields are as follows:

- Type:** record
- Operation:** create
- Decision Type:** Allow If
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details]
- Description:** (empty text area)
- Applies To:** No. of records matching the condition: 5. (Buttons: Add Filter Condition, Add OR Clause)
- Conditions:** Requires role (1 to 1 of 1). Role: bb2. (Buttons: Insert a new row...)
- Security Attribute Condition:** (empty text area)

At the bottom, a message states: 'No templates are available. Create A New One?'.

The screenshot shows the 'Institution Details' table in ServiceNow. The table has columns for Branch, Description, Email, Faculty Name, Phone Number, and Student Name. The data is as follows:

Branch	Description	Email	Faculty Name	Phone Number	Student Name
EEE		allan@gmail.com	Alfonso Griglen	989076	Allan Schwandt
EEE		allan@gmail.com	Alejandro Mascall	43543556	Allan Schwandt
EEE		alejandro@gmail.com	Abraham Lincoln	546738	Alejandro Mascall

On the right side, there is a user profile for 'EEE User' (Impersonating User) with links to Profile, Preferences, and Keyboard shortcuts.

Creation of Access Control Lists(Write ACL)

- 12. Click New to create a new ACL(Write)
- 13. Enter the following details:
 - Type: record
 - Operation: Write
 - Name: u_institution_details
 - Active: true
- 14. In the Requires role related list, add:
 - bb3 (custom role)
- 15. No need to add any Data condition
- 16. Save the ACL Record

servicenow

AllFavoritesHistoryWorkspacesAdmin

Access Controls

Search

UpdateDelete

Access Controlu_institution_details

* Type:record

* Operation:write

Decision Type:Allow If

Admin overrides☒

Protection policy:-- None --

* Name:Institution Details [u_institution_details]

Description:

Applies To:

Add Filter ConditionAdd OR Clause

Application:Global

Active☒

Advanced☐

Conditions

Requires role

Role

bb3

Security Attribute Condition

Local or Existing:ExistingLocal

Security Attribute:

Condition:

Data Condition

● User with **bb1 &bb2 &bb3** roles can view and edit the **EEE branch** records along with New Button

Institution Details Branch							Search	Actions on selected rows... New	
All									
☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number		
	CSE			Allie Pumphrey		Andrew Och	STU300		
	ECE			Abel Tuter		Aileen Mottern	STU100		
	EEE	ACL is excuted		Abraham Lincoln		Alejandro Mascall	STU200		

Creation of Access Control Lists(Delete ACL)

- 17. Click **New** to create a new ACL(Delete)
- 18. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
- 19. In the **Requires role** related list, add:
 - bb4 (custom role)
- 20. No need to add any Data condition
- 21. Save the ACL Record

servicenow

AllFavoritesHistoryWorkspacesAdmin

Access Controls

Search

UpdateDelete

Access Control

u_institution_details

* Type

record

* Operation

delete

Decision Type

Allow If

Admin overrides

☒

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Applies To

Add Filter Condition

Add OR Clause

Application

Global

Active

☒

Advanced

☐

Conditions

Requires role

Role

bb4

Security Attribute Condition

Local or Existing

☒ Existing ☐ Local

Security Attribute

Condition

Data Condition

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records**.

servicenow

AllFavoritesHistoryWorkspacesAdmin

Institution Details

Search

Actions on selected rows... New

All

Branch

Description

Email

Faculty Name

Phone Number

Student Name

Student Roll Number

CSE

Allie Pumphrey

Andrew Och

STU300

ECE

Abel Tutor

Aileen Mottern

STU100

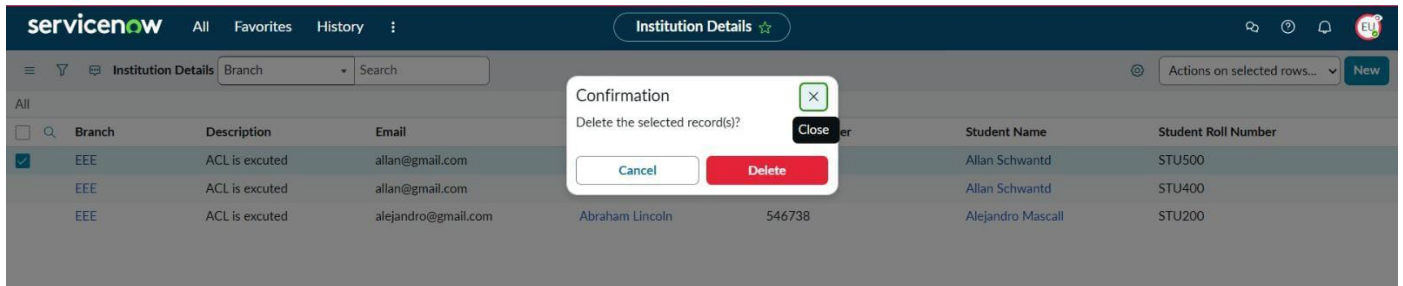
EEE

ACL is excuted

Abraham Lincoln

Alejandro Mascall

STU200



Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.